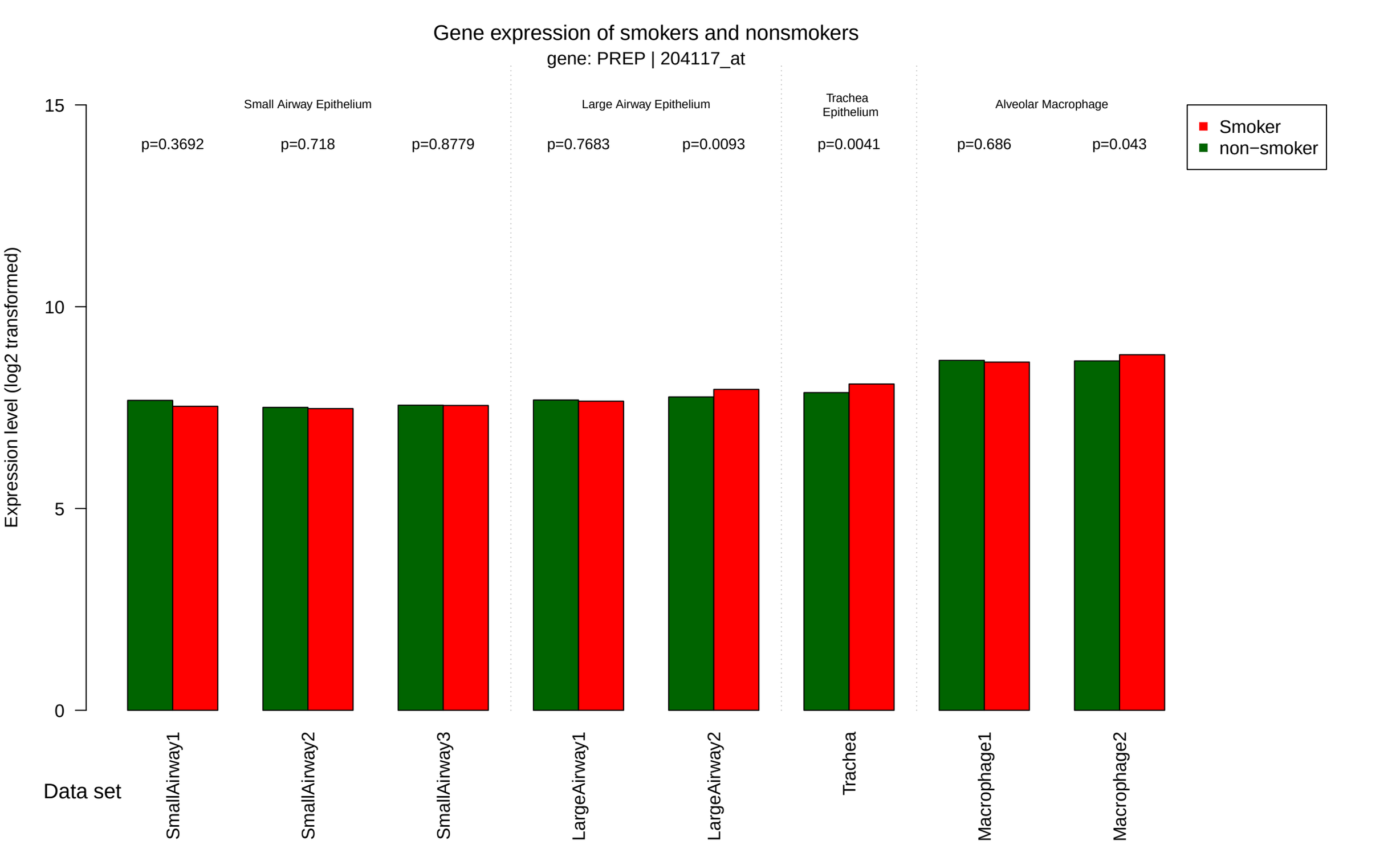
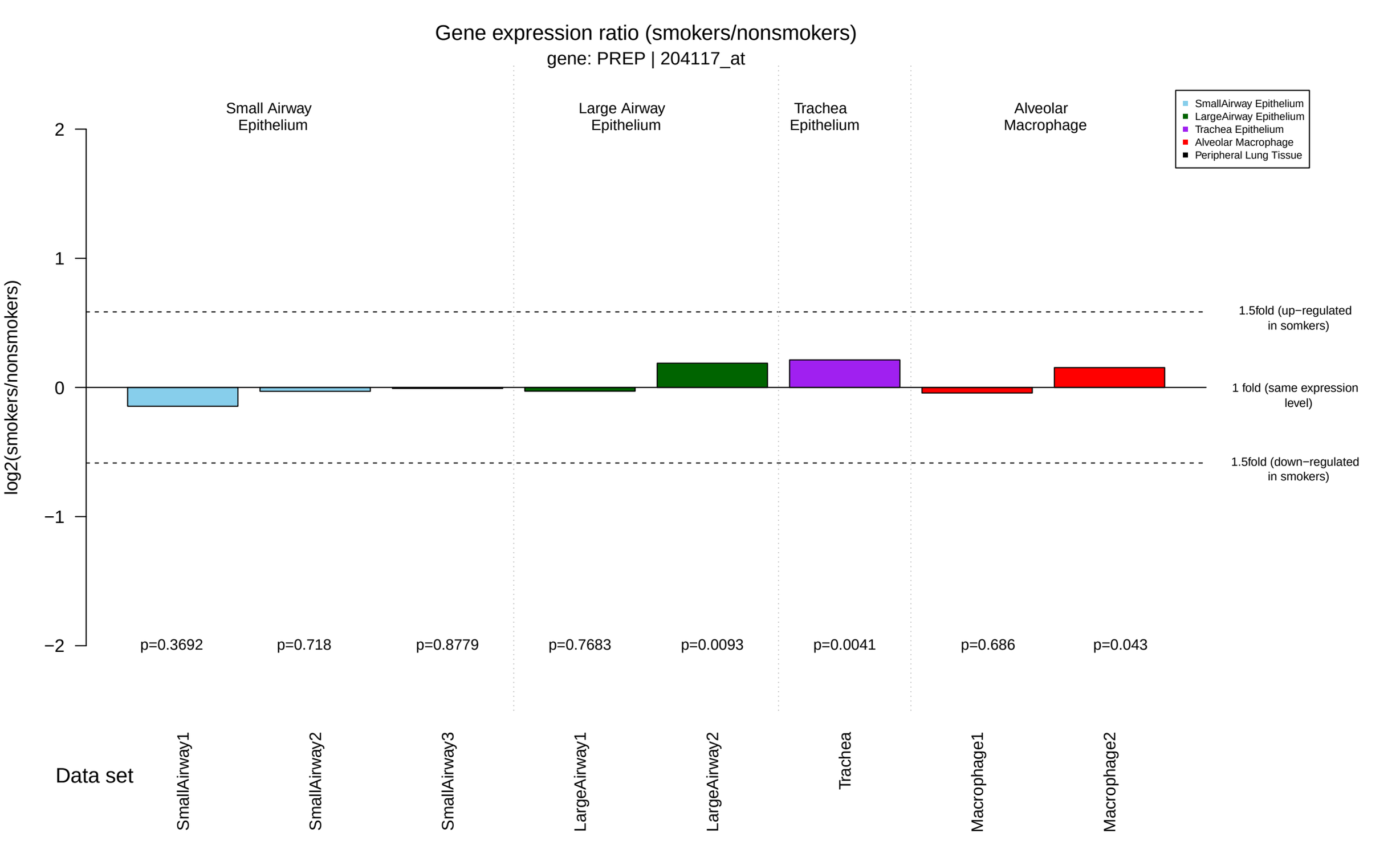
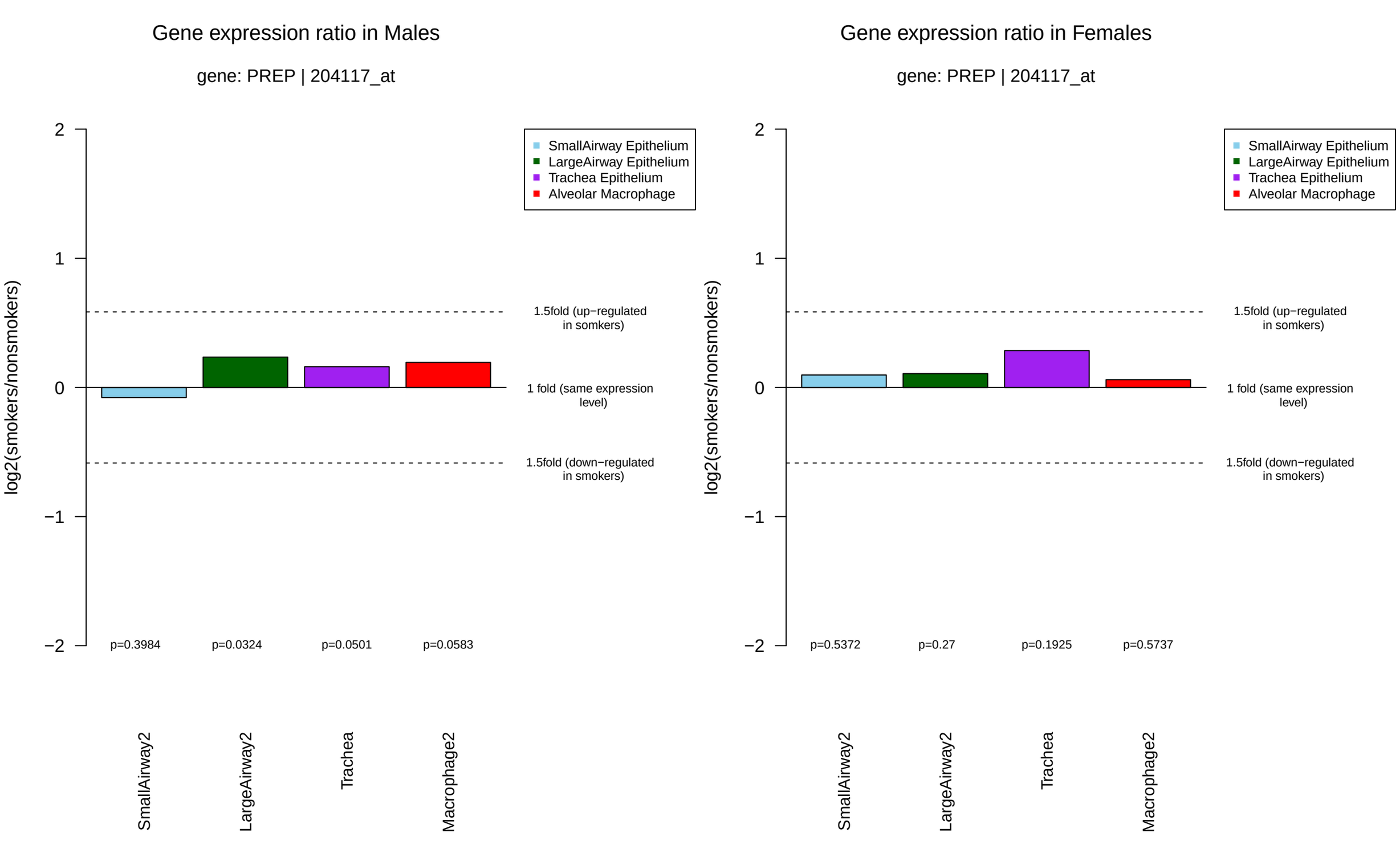
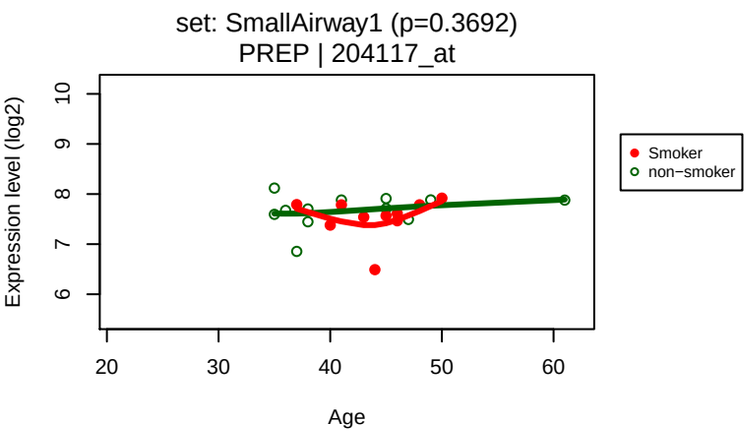
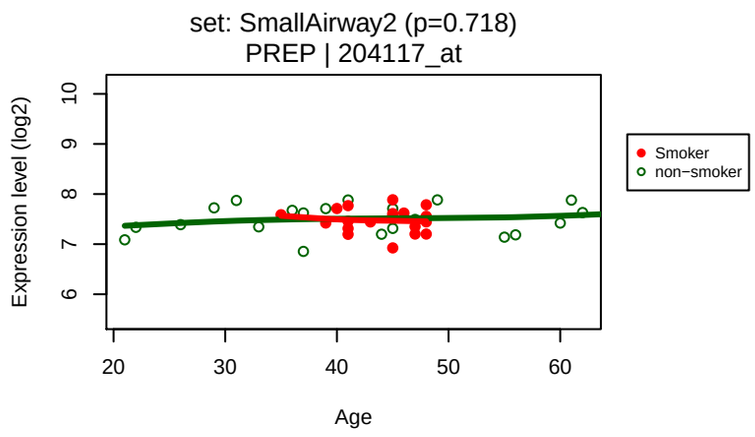
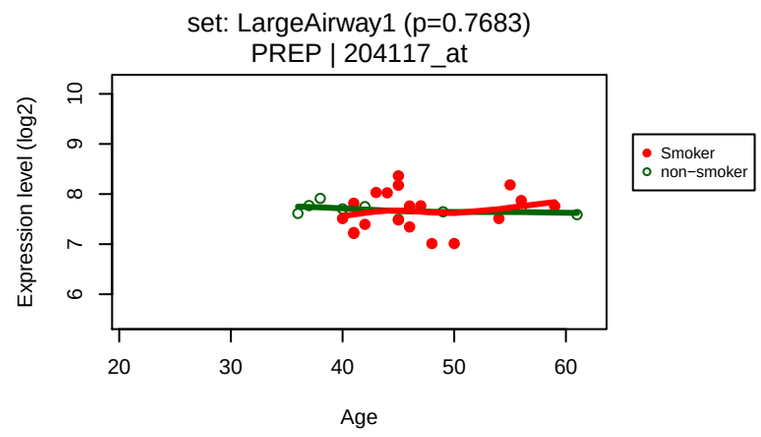

Gene expression of smokers and nonsmokers
gene: PREP | 204117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3692
p=0.718
p=0.8779
p=0.7683
p=0.0093
p=0.0041
p=0.686
p=0.043
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

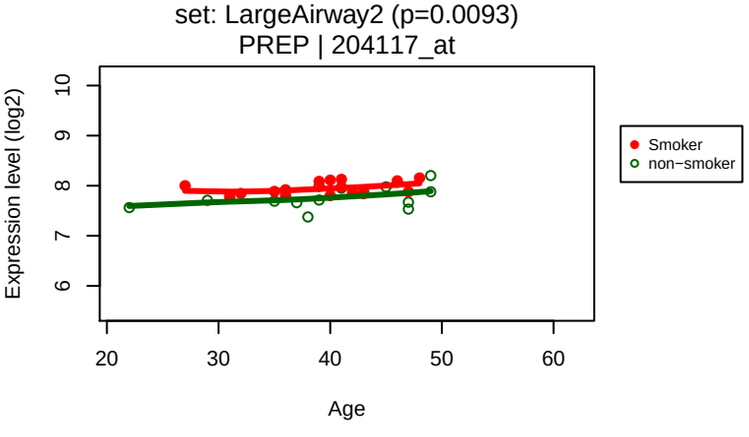

Gene expression ratio (smokers/nonsmokers)
gene: PREP | 204117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3692
p=0.718
p=0.8779
p=0.7683
p=0.0093
p=0.0041
p=0.686
p=0.043
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

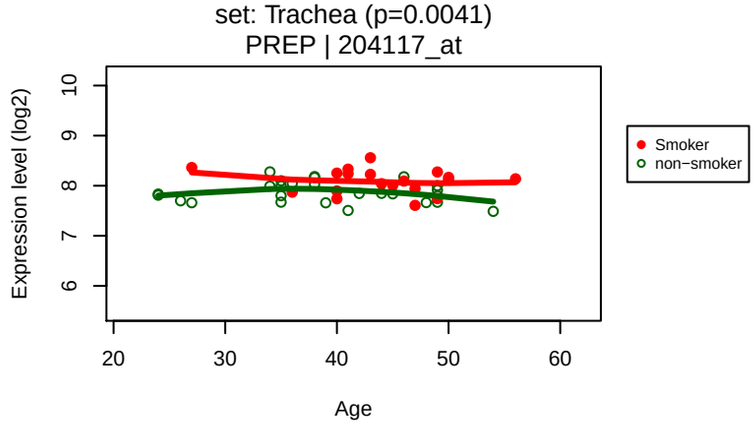

Gene expression ratio in Males
gene: PREP | 204117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3984
p=0.0324
p=0.0501
p=0.0583
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PREP | 204117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5372
p=0.27
p=0.1925
p=0.5737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

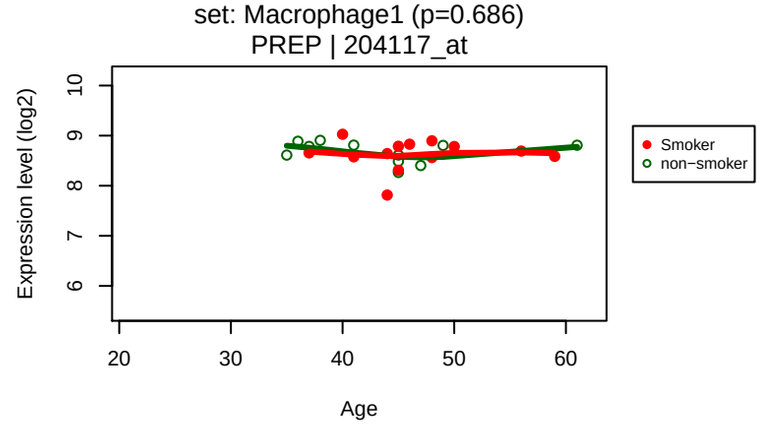

set: SmallAirway1 (p=0.3692)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker

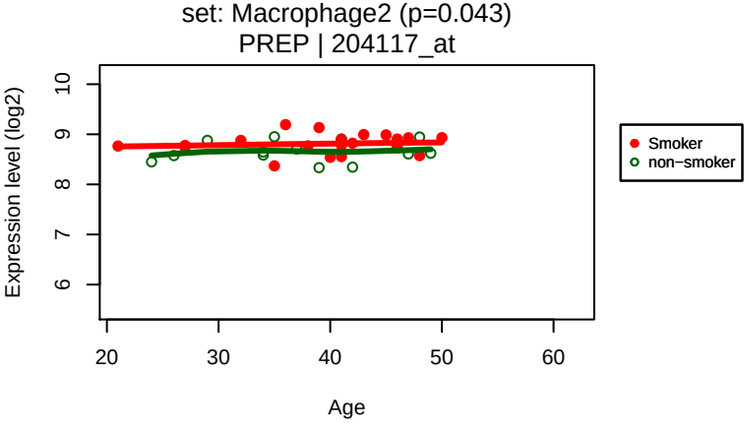

set: SmallAirway2 (p=0.718)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker

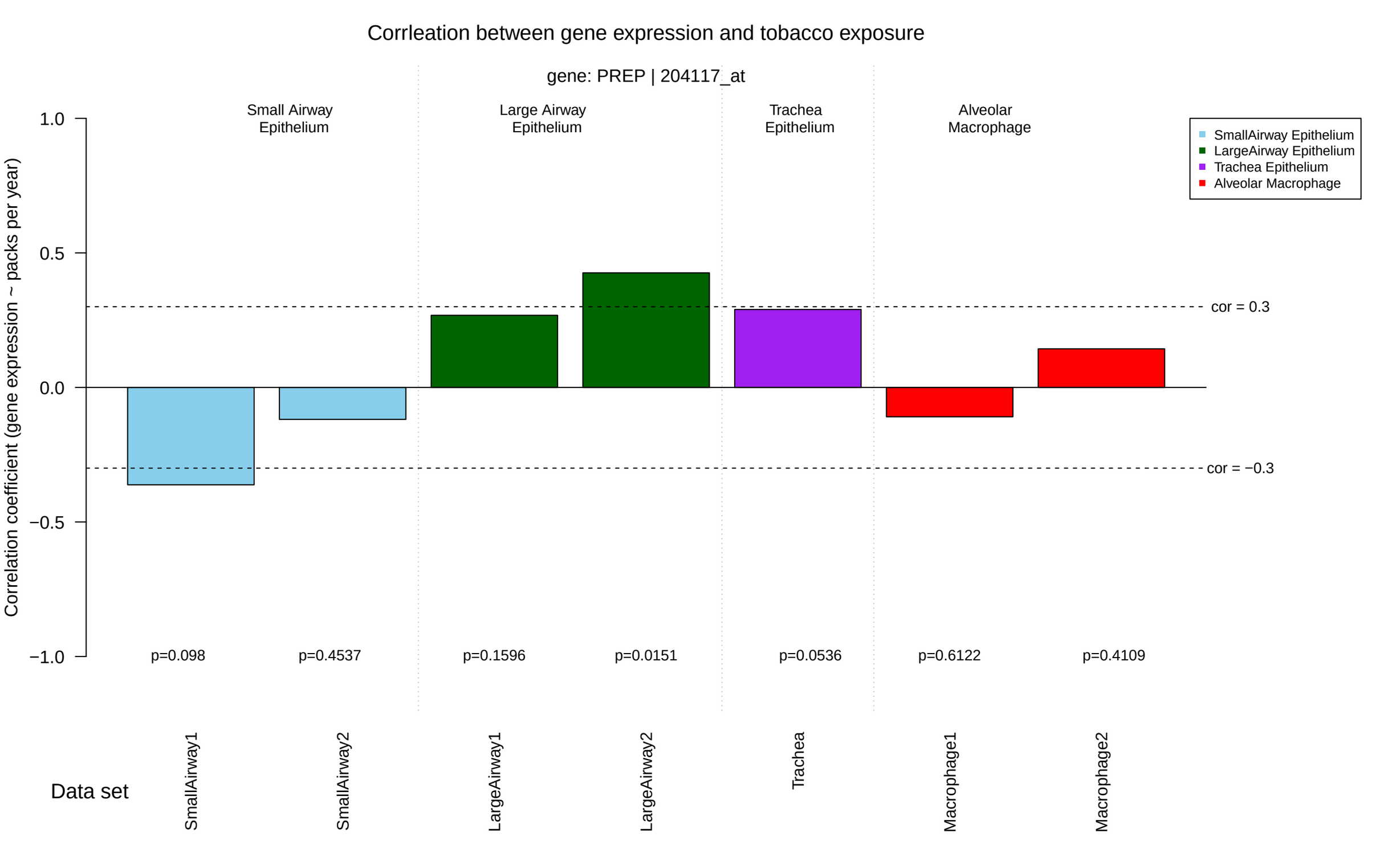

set: LargeAirway1 (p=0.7683)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0093)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0041)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.686)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.043)
PREP | 204117_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PREP | 204117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.098
p=0.4537
p=0.1596
p=0.0151
p=0.0536
p=0.6122
p=0.4109
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2